Pokémon Battle Prediction Report

FDS 2026 Kaggle Challenge, Sapienza University of Rome · Team PokéPredictors ·

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We predict whether Player 1 wins a Gen 2 OU battle from Player 1's team, Player 2's lead and the first 30 turns. The training set has 10,000 labelled battles (half of them wins) and the test set has 5,000. Our selected submission scored **0.69000 on the public leaderboard and 0.69457 on the private leaderboard**. Our key finding is that the battles are played by recurring players who reuse their teams, so a team's past results help predict the winner.

Validation strategy. We used five fold stratified cross validation on the 10,000 training battles and checked important changes on two more random splits. Features that use the labels are rebuilt inside every fold, so no battle ever sees its own result. We never tuned on the test set.

Feature engineering. Each battle has 589 features, computed for both players and as their difference: the state at turn 30 (HP left, faints, status conditions and the active Pokémon's boosts; "more HP left wins" alone is right 63% of the time), HP and faints at turns 5 to 25, move categories and super effective hits, the HP left of the 60 most common species, and the type advantage between the surviving Pokémon. Player 2 is often a player we see as Player 1 in other battles, so matching Player 2's revealed Pokémon to those full teams reveals the unseen ones in 72% of battles. Because Player 2's Pokémon appear only by name, we rebuilt the types and base stats of all 247 species from the competition files.

Player skill features. 65% of test battles reuse a team from the training set. Each team gets its win rate over its other battles, as Player 1 and as Player 2, smoothed towards 50% for teams with few games. We compute it in three variants: similar teams (five of six Pokémon shared), all possible Player 2 teams, and clutch skill (results compared with what the battle position predicts).

Models and refinement. For each variant we blend Logistic Regression (weight 0.3) with HistGradientBoosting (weight 0.7, depth 5, 600 trees), then average the three variants. Validation accuracy grew from 0.681 (battle state) to 0.687 (species and matchups), 0.691 (Player 1 skill), 0.698 (Player 2 matching) and 0.701 (final ensemble). An HP time series, move counts, team ratings, Random Forest, Extra Trees, CatBoost, XGBoost and blending with a second model built by our team gave no improvement.

Reproducibility. We use only classical models and the competition files. Our public Kaggle notebook clones our GitHub repository and reproduces the submission exactly. Twelve test battles are exact copies of training battles, and for those we use the known result.